Avionics Navigation Systems (ECE4635/ESE482)

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Sheet Two

Theoritical Questions:

Q1. Explain the purpose of the navigation equations in an aircraft navigation system. What types of sensor information are used by the navigation computer?

The navigation equations describe how the outputs of navigation sensors are processed by the navigation computer to determine the aircraft's:

- Position
- Velocity
- Altitude
- Heading
- Attitude

The navigation computer combines information from different sensors. Chapter 2 identifies three major types of information:

1. **Absolute position information** from radio aids, radar, and satellites.
2. **Dead-reckoning information** from inertial sensors, Doppler radar, and air-data sensors.
3. **Line-of-sight information** to stars or other references.

The navigation computer uses these measurements to estimate the aircraft's current state and calculate navigation quantities such as position, velocity, range, and course.

Q2. Why is the Earth modeled as an ellipsoid rather than as a perfect sphere in high-accuracy navigation systems?

The Earth is modeled as an **ellipsoid** rather than a perfect sphere because its actual shape is slightly flattened at the poles and wider around the equator.

The ellipsoidal model provides greater accuracy for navigation calculations, especially when calculating:

- Latitude and longitude changes
- Distances
- Position
- Velocity components
- Radii of curvature

For this reason, navigation systems use quantities such as the **meridian radius of curvature** R_M and **prime-vertical radius of curvature**

Q3. Define the following:

1. Meridian radius of curvature (R_M)
2. Prime-vertical radius of curvature (R_P)
3. Gaussian radius of curvature (R_G)

Explain why these quantities are important in navigation calculations.

1. Meridian radius of curvature R_M

R_M is the radius of the best-fitting circle in the **north-south vertical plane** at a particular point on the Earth's ellipsoid.

2. Prime-vertical radius of curvature R_P

R_P is the radius of the best-fitting circle in the **east-west vertical plane**.

3. Gaussian radius of curvature R_G

The Gaussian radius is defined as:

$$\boxed{R_G = \sqrt{R_M R_P}}$$

These radii are important because they relate the aircraft's **linear velocity components** to the corresponding **angular rates of latitude and longitude**.

Q4. Coordinate Frames

List and briefly explain the following coordinate frames used in navigation:

- Earth-Centered Earth-Fixed (ECEF)
- Earth-Centered Inertial (ECI)
- Geodetic spherical coordinates
- Geodetic wander-azimuth coordinates
- Direction cosines

1. Earth-Centered Earth-Fixed (ECEF)

An Earth-centered coordinate system whose axes rotate with the Earth.

It is useful for representing aircraft position relative to the Earth.

2. Earth-Centered Inertial (ECI)

An Earth-centered coordinate system that does **not rotate with the Earth**.

It is useful for inertial navigation and astronomical applications.

3. Geodetic Spherical Coordinates

These describe position using coordinates such as latitude, longitude, and radius.

4. Geodetic Wander-Azimuth Coordinates

These use a local horizontal coordinate system in which the azimuth reference can be allowed to "wander." They are useful in inertial navigation because they help avoid some problems associated with polar regions.

5. Direction Cosines

Direction cosines describe the orientation of one coordinate frame relative to another through the components of unit vectors.

They can be used to transform velocity or position information between coordinate systems.

Q5. Define dead reckoning (DR). Explain how aircraft position can be calculated from velocity measurements.

Dead reckoning (DR) is a navigation technique in which the aircraft's current position is calculated from a previously known position by using measurements of its velocity and integrating those velocities over time.

Velocity → Integration → Displacement → New Position

The major limitation is that errors in velocity measurements accumulate with time, causing the calculated position to drift from the true position.

Q6. Explain the difference between:

- True heading
- True track
- Ground speed
- Drift angle

Why can aircraft heading be different from ground-track direction?

True Heading

The direction in which the aircraft's longitudinal axis is pointing, measured relative to true north.

True Track

The actual direction in which the aircraft moves over the ground.

Ground Speed

The magnitude of the aircraft's velocity relative to the ground.

Drift Angle

The angle between the aircraft's heading and its ground-track direction.

When wind is present:

$$\boxed{\vec{V}_{ground} = \vec{V}_{air} + \vec{V}_{wind}}$$

Therefore, the aircraft may be pointing in one direction while moving over the ground in another direction.

Thus:

$$\boxed{\text{Heading} \neq \text{Ground Track}}$$

when crosswind produces drift.

Q7. Explain how wind affects the velocity of an aircraft relative to the ground.

Wind changes the aircraft's velocity relative to the ground.

The aircraft has a velocity relative to the air, represented by its airspeed and heading. The moving air mass itself has a wind velocity.

Therefore:

$$\boxed{\vec{V}_{ground} = \vec{V}_{air} + \vec{V}_{wind}}$$

For example, a crosswind can move the aircraft sideways relative to its intended path. As a result:

- Ground speed may increase or decrease.
- Ground-track direction may differ from heading.
- A drift angle is produced.

Q8. How can Doppler radar provide information useful for dead-reckoning navigation?

Explain the significance of:

- Ground speed
- Drift angle

Doppler radar can provide measurements of the aircraft's motion relative to the ground.

Doppler radar can measure: the **ground speed**, and the **drift angle**.

These measurements can then be used to determine the north and east components of ground velocity:

Therefore, Doppler radar provides direct information about the aircraft's ground motion, which is particularly useful for dead-reckoning calculations.

Q9. Earth Coordinates

Explain the relationship between north/east velocity components and changes in latitude and longitude.

The aircraft's horizontal velocity can be divided into two components: V_{north} and V_{east} The north component produces a change in **latitude**, while the east component produces a change in **longitude**.

The latitude rate is:

$$\boxed{\dot{\phi} = \frac{V_{north}}{R_M + h}}$$

The longitude rate is:

$$\boxed{\dot{\lambda} = \frac{V_{east}}{(R_P + h)\cos\phi}}$$

where:

- R_M= meridian radius of curvature
- R_P= prime-vertical radius of curvature
- h= aircraft altitude
- ϕ= latitude

Therefore: $\boxed{V_{north} \rightarrow \dot{\phi}}$

And $\boxed{V_{east} \rightarrow \dot{\lambda}}$

This is how linear aircraft velocity is converted into changes in geographic position.

Q10. Course Computation

What is the purpose of course computation in an aircraft navigation computer?

The purpose of **course computation** is to determine the required navigation information needed to guide the aircraft toward a desired waypoint or destination.

The navigation computer can calculate quantities such as:

- Range to the destination
- Bearing to the destination
- Desired course
- Relative bearing
- Course deviation

For example, in the local Cartesian approximation, the range is:

$$\boxed{D = \sqrt{(x - x_t)^2 + (y - y_t)^2}}$$

and the bearing can be calculated from the relative position of the aircraft and destination.

The calculated course information is then used by the aircraft's guidance and steering systems to help the aircraft reach the desired waypoint.

Practical Proplems:

Q11. An aircraft is flying at a horizontal speed of V=250 m/s with a true track angle of 30^0 . Calculate :

1. V north
2. Veast

$$V = 250\ m/s \qquad \psi = 30^\circ$$

1. North velocity

$$V_{north} = V\cos\psi \qquad V_{north} = 250\cos 30^\circ$$

$$\cos 30^\circ = 0.8660$$

$$V_{north} = 250(0.8660) \qquad \boxed{V_{north} = 216.5\ m/s}$$

2. East velocity

$$V_{east} = V\sin\psi$$

$$V_{east} = 250\sin 30^\circ \qquad V_{east} = 250(0.5)$$

$$\boxed{V_{east} = 125\ m/s}$$

Q12. An aircraft maintains constant velocity components Vnorth=180m/s, Veast=120m/s for 600s.

Calculate:

1. North displacement
2. East displacement
3. Total horizontal displacement

1. North displacement

$$\Delta North = V_{north}t \qquad \Delta North = 180(600)$$

$$\boxed{\Delta North = 108000\ m} \qquad \boxed{\Delta North = 108\ km}$$

2. East displacement

$$\Delta East = V_{east}t \qquad \Delta East = 120(600)$$

$$\boxed{\Delta East = 72000\ m} \qquad \boxed{\Delta East = 72\ km}$$

3. Total horizontal displacement

Using Pythagoras: $D = \sqrt{(\Delta North)^2 + (\Delta East)^2}$

$$D = \sqrt{108^2 + 72^2} \qquad D = \sqrt{11664 + 5184} \qquad D = \sqrt{16848} \qquad \boxed{D = 129.8\ km}$$

Q13. Dead-Reckoning Position

An aircraft starts at X0=20 km, Y0=50km and flies with V north=150 m/s, V east=100m/s for 10 minutes. Determine the new (x) and (y) coordinates.

Initial position: $x_0 = 20\ km$ $y_0 = 50\ km$

Velocity: $V_{north} = 150\ m/s$ $V_{east} = 100\ m/s$

Time: $t = 10\ min = 600\ s$

$$x = x_0 + \int V_{east} dt$$

$$y = y_0 + \int V_{north} dt$$

For constant velocity: $x = x_0 + V_{east} t$

$y = y_0 + V_{north} t$

East displacement $\Delta x = 100(600) = 60000\ m = 60\ km$

$x = 20 + 60$ $\boxed{x = 80\ km}$

North displacement $\Delta y = 150(600) = 90000\ m = 90\ km$

$y = 50 + 90$ $\boxed{y = 140\ km}$

Q14. Wind Components

An aircraft has VTAS =200m/s and true heading $\psi_T = 40°$, $\theta = 0°$, $\alpha = 0°$, $\beta = 0°$, The wind components are: V wind-north=20m/s, V wind-east=-30 m/s calculate V north and V east.

$V_{wind-north} = 20\ m/s$ $V_{wind-east} = -30\ m/s$

$$V_{north} = V_{TAS} \cos(\theta - \alpha) \cos(\psi_T + \beta) + V_{wind-north}$$

$$V_{east} = V_{TAS} \cos(\theta - \alpha) \sin(\psi_T + \beta) + V_{wind-east}$$

$\theta - \alpha = 0° - 0° = 0°$

$\psi_T + \beta = 40° + 0° = 40°$

$V_{north} = 200 \cos 0° \cos 40° + 20$

$\cos 0° = 1$

$\cos 40° = 0.7660$

$V_{north} = 200(1)(0.7660) + 20 = 153.21 + 20$

$\boxed{V_{north} = 173.21\ m/s}$

$V_{east} = 200 \cos 0° \sin 40° - 30$

$\sin 40° = 0.6428$

$V_{east} = 200(1)(0.6428) - 30 = 128.56 - 30$

$\boxed{V_{east} = 98.56\ m/s}$

Q15. Ground Speed and Ground Track

Using the results of Q14, calculate ground speed and determine the ground-track angle.

1. Ground speed

$$V_g = \sqrt{V_{north}^2 + V_{east}^2}$$

$$V_g = \sqrt{(173.21)^2 + (98.56)^2} \qquad V_g = \sqrt{29999.7 + 9713.6}$$

$$V_g = \sqrt{39713.3} \qquad \boxed{V_g \approx 199.3\ m/s}$$

2. Ground-track angle

$$\psi_g = \tan^{-1}\left(\frac{V_{east}}{V_{north}}\right)$$

$$\psi_g = \tan^{-1}\left(\frac{98.56}{173.21}\right) \qquad \psi_g = \tan^{-1}(0.5689) \qquad \boxed{\psi_g \approx 29.64^\circ}$$

Q16. Doppler Radar

A Doppler radar measures:

$V_g = 220\ m/s$, $\delta = 8°$ and Ground-track angle $\psi_g = 50°$. calculate the north and east components of ground velocity.

1. North velocity

$$V_{north} = V_g \cos(\psi_g + \delta)$$

$$\psi_g + \delta = 50^\circ + 8^\circ \quad = 58^\circ$$

$$V_{north} = 220 \cos 58^\circ \qquad \cos 58^\circ \approx 0.5299$$

$$V_{north} = 220(0.5299) \qquad \boxed{V_{north} \approx 116.6\ m/s}$$

2. East velocity

$$V_{east} = 220 \sin 58^\circ \qquad \sin 58^\circ \approx 0.8480$$

$$V_{east} = 220(0.8480) \qquad \boxed{V_{east} \approx 186.6\ m/s}$$

Q17. Drift Interpretation

An aircraft has a true heading of $\psi_T = 090°$, but its measured ground track is $\psi_g = 100°$ Determine:

1. The magnitude of the drift angle.
2. Whether the aircraft is being displaced toward the north or south relative to its intended eastward direction.

Explain your answer.

1. Drift angle

The difference is: $\delta = 100° - 90°$ $\boxed{\delta = 10°}$

2. Direction of displacement

A heading of: $090°$ means **due east**.

A ground track of: $100°$

is 10° clockwise from east, meaning the actual motion is toward the **southeast**.

Therefore, the aircraft is being displaced:

$\boxed{10° \text{ toward the south from its intended eastward direction}}$

Q18. Meridian Radius of Curvature

At latitude : $\phi = 30°$, $a = 6{,}378{,}137\ m$, $e^2 = 0.00669438$. Calculate (RM).

$$R_M = \frac{a(1-e^2)}{(1-e^2\sin^2\phi)^{3/2}}$$

$$\sin 30° = 0.5$$

$$\sin^2 30° = 0.25$$

$$e^2\sin^2\phi = 0.00669438(0.25) = 0.001673595$$

$$1 - e^2\sin^2\phi = 0.998326405$$

$$(0.998326405)^{3/2} \approx 0.997490$$

$$a(1-e^2) = 6,378,137(1-0.00669438) \approx 6,335,439\ m$$

$$R_M = \frac{6,335,439}{0.997490}$$

$\boxed{R_M \approx 6,351,377\ m}$ $\boxed{R_M \approx 6351.38\ km}$

Q19. Prime-Vertical Radius

At $\phi = 30°$, $a = 6{,}378{,}137\ m$, $e^2 = 0.00669438$. calculate the prime-vertical radius.

$$R_P = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}}$$

$$R_P = \frac{6,378,137}{0.999163} \qquad \boxed{R_P \approx 6,383,481\ m}$$

$$\boxed{R_P \approx 6383.48\ km}$$

Q20. Gaussian Radius

Using the values obtained in Q18 and Q19, calculate Gaussian Radius.

$$\boxed{R_G = \sqrt{R_M R_P}}$$

$$R_M = 6,351,377\ m$$

$$R_P = 6,383,481\ m$$

$$R_G = \sqrt{(6,351,377)(6,383,481)}$$

$$\boxed{R_G \approx 6,367,409\ m} \qquad \boxed{R_G \approx 6367.41\ km}$$

Q21. Latitude Rate

An aircraft is flying northward with $V_{north} = 250\ m/s$ at $\phi = 30°$ and altitude $h = 10{,}000\ m$
calculate the latitude rate in radians per second.

From Q18:

$$R_M = 6,351,377\ m$$

$$\boxed{\dot{\phi} = \frac{V_{north}}{R_M + h}}$$

$$\dot{\phi} = \frac{250}{6,351,377 + 10,000} \qquad \dot{\phi} = \frac{250}{6,361,377}$$

$$\boxed{\dot{\phi} = 3.93 \times 10^{-5}\ rad/s}$$

$$\dot{\phi} = 3.93 \times 10^{-5} \frac{180}{\pi} \qquad \boxed{\dot{\phi} \approx 0.002252°/s}$$

Q22. Longitude Rate

An aircraft has $V_{east} = 300\ m/s$, $\phi = 45°$, $h = 10{,}000\ m$, calculate the longitude rate.

From Q19: $R_P = 6{,}383{,}481\ m$

$$\boxed{\dot{\lambda} = \frac{V_{east}}{(R_P + h)\cos\phi}}$$

$$\dot{\lambda} = \frac{300}{(6,383,481 + 10,000)\cos 45°}$$

$$\boxed{\dot{\lambda} \approx 6.636 \times 10^{-5}\ rad/s} \qquad \boxed{\dot{\lambda} \approx 0.003802°/s}$$

Q23. Combined Position Update

An aircraft is at $\phi_0 = 30°$, with $\lambda_0 = 31°$, $V_{north} = 200\ m/s$, $V_{east} = 150\ m/s$, Calculate the approximate change in latitude and longitude after 600 s.

The question does **not specify altitude** h. For this solution, I will assume: $\boxed{h = 0}$ and use the $R_M$and $R_P$values calculated at 30°: $R_M = 6{,}351{,}377\ m$, $R_P = 6{,}383{,}481\ m$.

Latitude rate

$$\dot{\phi} = \frac{200}{6,351,377} \qquad \dot{\phi} = 3.1499 \times 10^{-5}\ rad/s$$

$$\Delta\phi = \dot{\phi}(600) \qquad \Delta\phi = 0.018899\ rad$$

$$\Delta\phi = 0.018899\frac{180}{\pi} \qquad \boxed{\Delta\phi \approx 1.083°}$$

$$\phi_f = 30 + 1.083 \qquad \boxed{\phi_f \approx 31.083°}$$

Longitude rate

$$\dot{\lambda} = \frac{150}{6,383,481 \cos 30^\circ}$$

$$\dot{\lambda} \approx 2.712 \times 10^{-5}\ rad/s$$

$$\Delta\lambda = \dot{\lambda}(600)$$

$$\Delta\lambda \approx 0.016276\ rad \qquad \boxed{\Delta\lambda \approx 0.933^\circ}$$

$$\lambda_f = 31 + 0.933$$

$$\boxed{\lambda_f \approx 31.933^\circ}$$

Q24. Range to a Waypoint

An aircraft is located at Aircraft at $(x,y) = (20,30)\ km$ and Destination at $(x_t,y_t) = (100,90)\ km$. Calculate the range.

$$D = \sqrt{(x - x_t)^2 + (y - y_t)^2}$$

$$D = \sqrt{(20 - 100)^2 + (30 - 90)^2}$$

$$D = \sqrt{(-80)^2 + (-60)^2}$$

$$D = \sqrt{6400 + 3600}$$

$$D = \sqrt{10000}$$

$$\boxed{D = 100\ km}$$

Q25. Bearing to a Waypoint

Using the same aircraft and waypoint positions from Q24, calculate the bearing .

$$B_r = \tan^{-1}\left(\frac{x - x_t}{y - y_t}\right)$$

$$x - x_t = -80$$
$$y - y_t = -60$$

$$B_r = \tan^{-1}\left(\frac{20 - 100}{30 - 90}\right)$$

$$B_r = \tan^{-1}\left(\frac{-80}{-60}\right) \qquad \boxed{B_r \approx 53.13^\circ}$$

Q26. Plane-Triangle Approximation

The aircraft is 500 nautical miles from a destination at an azimuth of (Br =45◦).

Calculate the spherical-Earth correction term $R_G = 3440\ nm$, For $\Phi = 30°$.

$$\Delta D = \frac{D^2}{4R_G} \sin B_r \sin 2B_r \tan \Phi$$

$$D^2 = 500^2 = 250000$$

$$4R_G = 4(3440) = 13760$$

$$\frac{D^2}{4R_G} = \frac{250000}{13760} \approx 18.1686$$

$$\Delta D = 18.1686(0.7071)(1)(0.57735)$$

$$\boxed{\Delta D \approx 7.42\ nm}$$

Q27. Complete Dead-Reckoning Problem

An aircraft has $V_{TAS} = 220\ m/s$

$$\psi_T = 60°$$
$$\theta = 2°$$
$$\alpha = 4°$$
$$\beta = 1°$$

The wind components are $V_{wind-north} = 25\ m/s$, $V_{wind-east} = -15\ m/s$, Time , The aircraft flies for 900 s.

Calculate:

1. V_north
2. V_east
3. North displacement
4. East displacement
5. Ground speed
6. Ground-track angle

1. Calculate V_{north}

$$\theta - \alpha = 2^\circ - 4^\circ = -2^\circ$$

$$\psi_T + \beta = 60^\circ + 1^\circ = 61^\circ$$

$$V_N = 220\cos(-2^\circ)\cos 61^\circ + 25$$

$$\cos(-2^\circ) = 0.99939$$

$$\cos 61^\circ = 0.48481$$

$$V_N \approx 220(0.99939)(0.48481) + 25$$

$$\boxed{V_N \approx 131.59\ m/s}$$

2. Calculate V_{east}

$$V_E = 220\cos(-2^\circ)\sin 61^\circ - 15$$

$$V_E \approx 220(0.99939)(0.87462) - 15$$

$$\boxed{V_E \approx 177.30\ m/s}$$

3. North displacement

$$\Delta N = V_N t$$

$$= 131.593(900)$$

$$= 118,433.8\ m \qquad \boxed{\Delta N \approx 118.43\ km}$$

4. East displacement

$$\Delta E = V_E t$$

$$= 177.299(900)$$

$$= 159,569.2\ m \qquad \boxed{\Delta E \approx 159.57\ km}$$

5. Ground speed

$$V_g = \sqrt{V_N^2 + V_E^2}$$

$$V_g = \sqrt{131.59^2 + 177.30^2} \qquad \boxed{V_g \approx 220.80\ m/s}$$

6. Ground-track angle

$$\psi_g = \tan^{-1}\left(\frac{V_E}{V_N}\right)$$

$$\psi_g = \tan^{-1}\left(\frac{177.30}{131.59}\right) \qquad \boxed{\psi_g \approx 53.42^\circ}$$

Q28. Doppler-Based Dead Reckoning

A Doppler radar provides $V_g = 250\ m/s$, $\delta = 6°$and the aircraft ground-track angle is $\psi_g = 75°$. The aircraft starts from (X0=0, Y0=0) and maintains the measured velocity for 20 minutes.

Determine:

1. North velocity
2. East velocity
3. North displacement
4. East displacement
5. Final dead-reckoned position

$$\psi_g + \delta = 75^\circ + 6^\circ = 81^\circ$$

1. North velocity

$$V_N = V_g \cos(\psi_g + \delta)$$

$$V_N = 250 \cos 81^\circ$$

$$\boxed{V_N \approx 39.11\ m/s}$$

2. East velocity

$$V_E = V_g \sin(\psi_g + \delta)$$

$$V_E = 250 \sin 81^\circ$$

$$\boxed{V_E \approx 246.92\ m/s}$$

3. North displacement

$$\Delta N = 39.1086(1200)$$

$$= 46,930.3\ m$$

$$\boxed{\Delta N = 46.93\ km}$$

4. East displacement

$$\Delta E = 246.922(1200)$$

$$= 296,306.5\ m$$

$$\boxed{\Delta E = 296.31\ km}$$

5. Final dead-reckoned position

$$x_0 = 0, \qquad y_0 = 0$$

$$\boxed{x_f = 296.31\ km}$$

$$\boxed{y_f = 46.93\ km}$$

So the aircraft finishes approximately:

296.31 km east and 46.93 km north of its starting point.

Q29. Airspeed vs Ground Velocity

An aircraft flies with $V_{TAS} = 250\ m/s.$ toward $90°$. A wind of $50\ m/s$ is blowing toward $180°$. Determine:

1. North wind component
2. East wind component
3. Ground north velocity
4. Ground east velocity
5. Ground speed
6. Ground-track direction
7. Drift angle

Explain physically why the ground track is not equal to the aircraft heading.

1. North wind component

A wind toward 180° is directed **south**.

Therefore: $\boxed{V_{wind-N} = -50\ m/s}$

2. East wind component

There is no east/west component: $\boxed{V_{wind-E} = 0}$

3. Ground north velocity

Aircraft is flying east, so its north component is zero.

Therefore: $V_N = 0 + (-50)$

$$\boxed{V_N = -50\ m/s}$$

4. Ground east velocity

The aircraft has: $V_E = 250\ m/s$

and the wind has zero east component.

Therefore: $\boxed{V_E = 250\ m/s}$

5. Ground speed

$$V_g = \sqrt{(-50)^2 + 250^2}$$

$$V_g = \sqrt{2500 + 62500}$$

$$V_g = \sqrt{65000}$$

$$\boxed{V_g \approx 254.95\ m/s}$$

6. Ground-track direction

Because the aircraft is moving east and south:

$$\psi_g = \tan^{-1}\left(\frac{250}{-50}\right)$$

The correct quadrant must be considered.

Therefore: $\boxed{\psi_g \approx 101.31^\circ}$

That means the aircraft is moving **10?** Wait: 101.31° is 11.31° south of east.

7. Drift angle

Heading: $\psi_T = 90^\circ$

Ground track: $\psi_g = 101.31^\circ$

Therefore: $\delta = 101.31 - 90$

$\boxed{\delta \approx 11.31^\circ}$

Physical explanation

The aircraft points toward east, but the wind pushes it toward the south.

Therefore: $\boxed{\text{Heading} \neq \text{Ground Track}}$

The aircraft is pointing east but actually moving toward the **southeast**.

Q30. Navigation Computer Problem

A navigation computer receives the following information:

- True airspeed
- Pitch angle
- Angle of attack
- Sideslip angle
- True heading
- Wind north component
- Wind east component

Write the sequence of calculations required to obtain the aircraft's ground velocity and then its dead-reckoned position.

The calculation can be divided into **four main stages**.

Step 1 – Calculate north velocity

Use: $$\boxed{V_N = V_{TAS}\cos(\theta - \alpha)\cos(\psi_T + \beta) + V_{wind-N}}$$

Step 2 – Calculate east velocity

$$\boxed{V_E = V_{TAS}\cos(\theta - \alpha)\sin(\psi_T + \beta) + V_{wind-E}}$$

These equations convert the aircraft's air-relative velocity and wind into its ground-relative velocity components.

Step 3 – Integrate velocity to obtain displacement

For the simple flat-Earth approximation:

$$\boxed{\Delta N = \int V_N dt}$$

$$\boxed{\Delta E = \int V_E dt}$$

If the velocity is constant:

$$\Delta N = V_N t$$

$$\Delta E = V_E t$$

Step 4 – Update aircraft position

If the initial position is: (N_0, E_0)

then:

$$\boxed{N = N_0 + \Delta N}$$

$$\boxed{E = E_0 + \Delta E}$$

For actual Earth navigation, the linear north/east velocities are converted to latitude and longitude rates:

$$\boxed{\dot{\phi} = \frac{V_N}{R_M + h}}$$

$$\boxed{\dot{\lambda} = \frac{V_E}{(R_P + h)\cos\phi}}$$

Thus the overall process is:

$$\boxed{\text{Aircraft/Air Data} \rightarrow V_N, V_E \rightarrow \text{Integration} \rightarrow \text{Position}}$$

Part J – Higher-Level Conceptual / Discussion Questions

Q31. Why does the dead-reckoning calculation use velocity integration rather than simply multiplying airspeed by time?

The flat-Earth approximation treats the Earth locally as a plane.

This can be useful for:

- Short distances
- Small areas
- Simple calculations

However, over long distances the Earth's curvature becomes significant.

A high-accuracy navigation system must account for:

- Earth's ellipsoidal shape
- Latitude
- Longitude
- R_M
- R_P
- R_G

For example:

$$\dot{\phi} = \frac{V_N}{R_M + h}$$

and:

$$\dot{\lambda} = \frac{V_E}{(R_P + h)\cos\phi}$$

Thus the Earth model becomes an important part of accurate navigation.

Q32. Why is the flat-Earth approximation acceptable for some short-distance calculations but inappropriate for high-accuracy global navigation?

The length of one degree of longitude depends on latitude.

At the equator, lines of longitude are farthest apart.

As we move toward the poles, lines of longitude become closer together.

The east-west distance corresponding to a change in longitude is therefore proportional to:

$$\cos\phi$$

This is why: $\boxed{\dot{\lambda} = \frac{V_E}{(R_P+h)\cos\phi}}$

contains $\cos\phi$.

As: $\phi \to 90°$

then: $\cos\phi \to 0$

which explains why longitude becomes increasingly problematic near the poles.

Q33. Complete Navigation Solution

An aircraft is initially at $\phi_0 = 30°$, $\lambda_0 = 31°$, Altitude $h = 8000\ m$, The aircraft parameters are $V_{TAS} = 230\ m/s$, $\theta = 2°$, $\alpha = 3°$, $\beta = 2°$, $\psi_T = 45°$. Flight time t=1200 s.
Determine:

1. Rm
2. Rp
3. V north
4. V east
5. Ground speed
6. Ground-track direction
7. $\dot{\phi}$
8. $\dot{\lambda}$
9. Approximate change in latitude
10. Approximate change in longitude
11. Final approximate latitude and longitude

Finally, explain which assumptions are made when integrating the velocity components over the 1200-second interval.

$$V_{wind-N} = 20\ m/s$$

$$V_{wind-E} = 30\ m/s$$

$$t = 1200\ s$$

1. Calculate R_M

At: $\phi = 30^\circ$

$$\boxed{R_M = 6,351,377\ m}$$

2. Calculate R_P

$$\boxed{R_P = 6,383,481\ m}$$

3. Calculate V_{north}

$$V_N = V_{TAS}\cos(\theta - \alpha)\cos(\psi_T + \beta) + V_{wind-N}$$

$$\theta - \alpha = 2^\circ - 3^\circ = -1^\circ$$

3. Calculate V_{north}

$$V_N = V_{TAS}\cos(\theta - \alpha)\cos(\psi_T + \beta) + V_{wind-N}$$

$$\theta - \alpha = 2^\circ - 3^\circ = -1^\circ$$

$$\psi_T + \beta = 45^\circ + 2^\circ = 47^\circ$$

$$V_N = 230\cos(-1^\circ)\cos 47^\circ + 20$$

$$\boxed{V_N \approx 176.84\ m/s}$$

4. Calculate V_{east}

$$V_E = 230\cos(-1^\circ)\sin 47^\circ + 30$$

$$\boxed{V_E \approx 198.19\ m/s}$$

5. Ground speed

$$V_g = \sqrt{V_N^2 + V_E^2}$$

$$= \sqrt{176.84^2 + 198.19^2}$$

$$\boxed{V_g \approx 265.61\ m/s}$$

6. Ground-track direction

$$\psi_g = \tan^{-1}\left(\frac{V_E}{V_N}\right)$$

$$= \tan^{-1}\left(\frac{198.19}{176.84}\right)$$

$$\boxed{\psi_g \approx 48.26^\circ}$$

7. Latitude rate

$$\dot{\phi} = \frac{V_N}{R_M + h}$$

$$= \frac{176.836}{6,351,377 + 8,000}$$

$$\boxed{\dot{\phi} \approx 2.781 \times 10^{-5}\ rad/s}$$

8. Longitude rate

$$\dot{\lambda} = \frac{V_E}{(R_P + h)\cos\phi}$$

$$= \frac{198.186}{(6,383,481 + 8,000)\cos 30^\circ}$$

$$\boxed{\dot{\lambda} \approx 3.580 \times 10^{-5}\ rad/s}$$

9. Change in latitude

$$t = 1200\ s$$

$$\Delta\phi = \dot{\phi}t$$

$$= (2.781 \times 10^{-5})(1200)$$

$$= 0.03337\ rad$$

$$\boxed{\Delta\phi \approx 1.912^\circ}$$

$$\phi_f = 30 + 1.912$$

$$\boxed{\phi_f \approx 31.912^\circ}$$

10. Change in longitude

$$\Delta\lambda = \dot{\lambda}t$$

$$= (3.580 \times 10^{-5})(1200)$$

$$= 0.04297\ rad$$

$$\boxed{\Delta\lambda \approx 2.462^\circ}$$

$$\lambda_f = 31 + 2.462$$

$$\boxed{\lambda_f \approx 33.462^\circ}$$

11. Final Position

Therefore, using the constant-rate approximation:

$$\boxed{\phi_f \approx 31.912^\circ}$$

$$\boxed{\lambda_f \approx 33.462^\circ}$$

Q34. The aircraft in Q59 is flying near (80^0) north latitude instead of (30^0).

Without performing the complete calculation, discuss how and why the relationship between eastward velocity and longitude rate changes.

Explain the significance of the (cos) term.

Answer

The longitude-rate equation is: $\boxed{\dot{\lambda} = \frac{V_E}{(R_P+h)\cos\phi}}$

At: $\phi = 30^\circ$ we have $\cos 30^\circ = 0.866$

But at: $\phi = 80°$ we have $\cos 80° \approx 0.174$
Therefore, the denominator becomes much smaller.

For the **same eastward velocity** V_E: $\boxed{\dot{\lambda} \text{ becomes much larger}}$

Why?

As the aircraft approaches the North Pole, the meridians become increasingly close together. Therefore, a relatively small eastward physical displacement corresponds to a relatively large change in longitude.

Conceptually: $\boxed{\text{High latitude} \rightarrow \cos\phi \downarrow \rightarrow \dot{\lambda} \uparrow}$

For example, compare: $\frac{1}{\cos 30°} \approx 1.155$

with: $\frac{1}{\cos 80°} \approx 5.759$

So, ignoring the small change in R_P, the longitude rate for the same eastward velocity is roughly **5 times larger at** $80°$than at $30°$.

This is one reason why longitude becomes a particularly sensitive coordinate near the poles.

$$\boxed{\dot{\lambda} \propto \frac{1}{\cos\phi}}$$

Therefore:

- Near the equator → longitude changes relatively slowly for a given eastward velocity.
- Near the poles → longitude changes very rapidly for the same eastward velocity.

And as:

$$\phi \rightarrow 90°$$
$$\cos\phi \rightarrow 0$$

so the longitude coordinate becomes **singular at the pole**. This is also why navigation systems need appropriate coordinate-frame strategies in polar regions.

PROBLEMS

2.1. The direction cosine matrix $[C]$ transforms the Earth-centered inertial coordinates y_i into the locally-level navigation coordinates z_j. Let $\alpha = 0$ when the aircraft is on the equator, and let the initial matrix be

$$[C] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Let $\dot{\alpha} = \dot{\lambda}$.

(a) If the aircraft flies 90° due east, show the direction cosine matrix.

$$Ans. \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

(b) If the aircraft flies 90° due north from its original position, show the direction cosine matrix.

Ans:

$$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) If the aircraft flies 30° due east on the equator at 600 knots from its original position, what are the C_{ij} and the $\dot{C}_{ij}$? Use Equation 7.40.

Ans:

$$[\dot{C}] = -\frac{5}{57.3} \begin{bmatrix} 1 & \sqrt{3} & -\sqrt{3} \\ -\sqrt{3} & 1 & 1 \\ 1 & -\sqrt{3} & 0 \end{bmatrix} \text{hr}^{-1}$$

2.2. An aircraft flies 3 hrs east then 2 hrs north at 300 knots at an altitude of 3 nmi, starting at 40° north latitude.

(a) Find its position using the flat-Earth dead-reckoning equations.

Ans. $x = 900$ nmi, $y = 600$ nmi.

(b) Find the final latitude-longitude using spherical-Earth equations with the Gaussian radius of curvature.

Ans. $\Phi = 49.98°$, $\Delta\lambda = 19.54°$.

(c) Find the final latitude-longitude using the ellipsoidal-Earth equations, (2.5).

Ans. $\Phi = 49.99°$, $\Delta\lambda = 19.50°$.

(d) Find the distance from the start to the destination of case **a** using the flat-Earth range equation, 2.20.

Ans. 1081.7 nmi.

(e) Find the distance to the destination of case **b** using the spherical-Earth approximation with the Gaussian radius of curvature.

Ans. 1022.7 nmi.

(f) Find the distance to the destination of case **b** using the flat-Earth equations plus the correction of Equation 2.21.

Ans. 1027.0 nmi.

2.3. A GPS satellite is at the ECEF coordinates:

$$x = 14367.71 \text{ nmi}, \quad y = 0, \quad z = 0$$

The observer is at latitude 45°, longitude 45°, 30,000-ft altitude. Calculate the distance from the aircraft to the satellite and the elevation of the line of sight. Use the WGS dimensions on page 25.

Ans. $R = 12{,}986.56$ nmi, $\theta = 43.5$ deg.

2.4. Derive Equation 2.21. *Hint:* solve the spherical triangle for the cosine of the range angle and express the coordinates of the waypoint as the present position plus a small increment. Expand the sines to third order and the cosines to second order.

2.5. Verify the calculations on page 46 for the probability of falling inside a rectangle of edge 1.2 nmi ($x = 0.6$ nmi) by 2.4 nmi ($y = 1.2$ nmi). Let $\sigma_x = 0.4$ nmi and $\sigma_y = 2.4$ nmi.